

Deepesh Agrawal

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Summary

AI architect & ML engineer with **9+ years** designing production-grade solutions across NLP, CV and Generative AI. Proven track record of **shipping scalable agent platforms, RAG systems and MLOps pipelines** that cut latency, boost accuracy and save engineer hours. Experienced tech-lead who has driven cross-functional teams, audits and client deliveries for IBM, Fujitsu and Fortune 500 customers.

Experience

Senior AI Engineer | Presight AI, Abu Dhabi, UAE | May 2026 - Present

- Built an **audio analytics pipeline** for Presight Synergy to identify **who spoke, when, and what was said** by combining **pyannote.audio** speaker diarization with **Whisper large-v3** speech-to-text, while evaluating **Microsoft VibeVoice** for long-form, multi-speaker conversational audio workflows.
- Worked on **video analysis and signature detection** for Synergy use cases using **Qwen-family multimodal / vision-language models**, applying **VLM prompting, frame-level analysis, structured detection outputs, and agentic AI workflow patterns**.
- Used **Ray Serve, NVIDIA Triton Inference Server, and vLLM** for scalable model serving across **LLM, VLM, speech, and multimodal workloads**, supporting **MCP-ready / agentic AI integration** for exposing model inference and analytics workflows through conversational interfaces.

 **Technologies:** Python, Java, PyTorch, pyannote.audio, Whisper, Microsoft VibeVoice, Qwen-family VLMs, Ray Serve, NVIDIA Triton Inference Server, vLLM, LangChain, LangGraph, MCP, Vector DBs

AI Engineer Architect | IBM Labs, Bengaluru, India | May 2025 - Apr 2026

- Contributed to pretraining-related work and led the post-training, alignment, evaluation, and enterprise deployment of **Granite-4.0-H-Small** from Granite-4.0-H-Small-Base, using supervised fine-tuning (**SFT**), reinforcement-learning-based alignment (**RLA**), and model merging to improve instruction following and production readiness for large-scale agent systems.
- Re-architected production **LLM-based agent platform** for large-file workloads, reducing **memory usage by ~3×** and eliminating stability issues under load, enabling reliable enterprise-scale operations.
- Converted blocking data operations into **asynchronous** execution pipelines, improving platform scalability and supporting higher concurrent agent interactions.
- Designed **context-aware, multi-step agent** workflows using graph-based orchestration and vector retrieval, enhancing planning depth and response quality.
- Developed actionable agents via **MCP-based tool** integration and external APIs (e.g., GitHub), enabling reliable real-world system interaction.
- Improved LLM output robustness and alignment using advanced prompting techniques, including **Chain-of-Thought (CoT)** and **Tree-of-Thought (ToT)** reasoning, along with structured reasoning evaluation and cross-team model optimization, boosting reliability and production readiness.
- Enabled dynamic, user-specific registry and CRUD operations for agents, OpenAPI tools, and MCP servers within the **Agentic AI Framework (ARIA)**, supporting agent-to-agent communication and enhancing platform flexibility, customization, and real-time adaptability.

 **Technologies:** Python, watsonx.ai, OpenAI GPT-4o, LangChain, LangGraph, MCP, Kubernetes/OpenShift, Kafka, Redis, Vector DBs (Milvus/Chroma), Grafana, Terraform, pytest.

Senior ML Engineer | Fujitsu Research, Bengaluru, India | May 2023 - May 2025

- Shipped a cross-lingual **Retrieval-Augmented Generation (RAG) Q&A chatbot platform** serving 5 languages, boosting answer **F1 $\approx$ 32%** versus baseline; production REST API (Flask) delivers **p95 latency <140 ms** on ARM-based EC2. Implemented **domain adaptation** via **fine-tuning BGE-M3 embeddings** and **LLaMa 3.1 8B**, and integrated a **reranker** to improve retrieval relevance and answer accuracy across languages.
- **Built a hands-free MLOps pipeline (Jenkins $\rightarrow$ Kubeflow $\rightarrow$ MLflow)** that auto-trains, quantizes, Canary-deploys and Grafana-monitors models-cutting release cycles from **10 days to 3 days**.
- **Optimised oneDNN operators for aarch64 CPUs**, trimming end-to-end inference time **41 %** and lowering AWS compute cost $\approx$ 18 %.

🔗 **Technologies:** Python, C++, PyTorch, LangChain, Haystack, LlamaIndex, Generative AI, Jenkins, Git, MLflow, Kubeflow, Grafana, Prometheus, Apache Airflow, AWS, Azure, llama.cpp.

Data Scientist | UST (Abzooba India Infotech), Pune, India | 📅 May 2021 - May 2023

- **Fine-tuned T5-Large for natural-language** → **SQL** (Capital Group), reaching **91 % query accuracy** on production logs and powering **≈ 12 K self-serve queries/month**.
- **Automated table extraction from “messy” Excel reports** for Swiss Re with ML + OCR, slashing manual prep time **70 %** (8 hrs → 0.5 hrs per report) and boosting pipeline throughput **16×**.
- Built an **ML model for one-year diabetic risk prediction** using **two years of patient medical and health insurance data**, enabling proactive patient monitoring and early intervention strategies.

🔗 **Technologies:** Python, PyTorch, Hugging Face Transformers, Flask, Docker, MySQL, Generative AI (GenAI), LLMs, OpenPyXL, Pandas, scikit-learn, NLP, BitBucket.

ML Engineer | AMD India Pvt. Ltd. (Contract), Bengaluru, India | 📅 Feb 2020 - May 2021

- Integrated **ZenDNN (AMD's deep learning library)** with PyTorch & ONNX Runtime.
- Conducted **extensive benchmarking of ZenDNN vs. ONNX Runtime**, achieving a **25% speedup in inference on AMD EPYC CPUs**

🔗 **Technologies:** Python, PyTorch, ONNX Runtime, ZenDNN, Docker, PyTest.

Solution Architect - Deep Learning | BrainGrid Technologies, India | 📅 Sep 2019 - Feb 2020

- **Optimized Daimler's deep learning code** in collaboration with **NVIDIA engineers**.
- **Deployed YOLO-based object detection** models on Jetson Nano for **real-time video processing**.

🔗 **Technologies:** Python, CUDA, PyTorch, TensorRT, Jetson Nano, Nsight System, OpenCV.

Machine Learning Engineer | Beahead Pvt. Ltd., Noida, India | 📅 Sep 2018 - Sep 2019

- **Diabetic Retinopathy Detection:** Trained InceptionResNetV2 to classify retinal images.
- **Video Analysis for Social Media Content:** Built ML models for content moderation in social media videos, using CNN for image classification, LSTM for sentiment analysis, and speech recognition.
- **Social Media Data Scraping & Keyword Extraction:** Automated data scraping from Facebook, Instagram, and Twitter, extracting insights using RAKE for keyword extraction.
- **Text Classification:** Automated OCR-based document classification for pharma guidelines using NLP.

🔗 **Technologies:** Python, TensorFlow, Keras, CNN, RNN, LSTM, OpenCV, Flask, Docker, Azure, scikit-learn, Pandas, Selenium, BeautifulSoup, NLP, Textract, Tesseract, OCR, RAKE

AI Engineer | Iotechworld Avigation Pvt. Ltd., Gurgaon, India | 📅 Feb 2018 - June 2018

- **Hand Gesture-Controlled Drone:** Developed a gesture-controlled drone using an SSD-based deep learning model, deployed on Raspberry Pi 3.
- **Drone Object Tracking & Navigation:** Built an object tracking & navigation system using Mean Shift algorithm and GPS data for drone movement.

🔗 **Technologies:** Python, TensorFlow API, SSD, CNN, Raspberry Pi 3, OpenCV, GPS, Mean-Shift Algorithm

Technical Skills

🔗 **GenAI / Agentic AI:** RAG, LLM agents, tool integration (MCP), LangChain, LangGraph, LlamaIndex, embeddings, vector retrieval

🔗 **Machine Learning & AI:** PyTorch, TensorFlow, Hugging Face Transformers, Scikit-Learn, CNN, RNN, LSTM

🔗 **Programming Languages:** Python, C++, MATLAB, R, SQL, CUDA, RAPIDS

🔗 **MLOps & DevOps:** MLflow, Kubeflow, Airflow, Jenkins, Docker, Kubernetes, Grafana, Prometheus, Terraform

🔗 **Optimization & Inference:** TensorRT, vLLM, oneDNN, ONNX Runtime, CuDNN, CuBLAS, llama.cpp

🔗 **Cloud Platforms:** AWS, Azure, Google Cloud

🔗 **Databases & Vector Search:** MySQL, PostgreSQL, Cassandra, FAISS, ChromaDB, qdrant

Education

🎓 **Master of Technology (MTech) in AI & ML 2023 - 2025**

📍 **BITS (Birla Institute of Technology and Science), Pilani, India**

Certifications

- **Software Architecture & Design of Modern Large-Scale Systems** - Udemy
- **Introduction to Generative AI** - Google Cloud
- **Microsoft DP-100: Designing & Implementing Data Science Solutions on Azure**
- **Advanced Data Science** - IBM
- **Computer Vision with Embedded Machine Learning** - Edge Impulse
- **Neural Networks & Deep Learning** - DeepLearning.ai
- **Machine Learning on Google Cloud** - Google Cloud